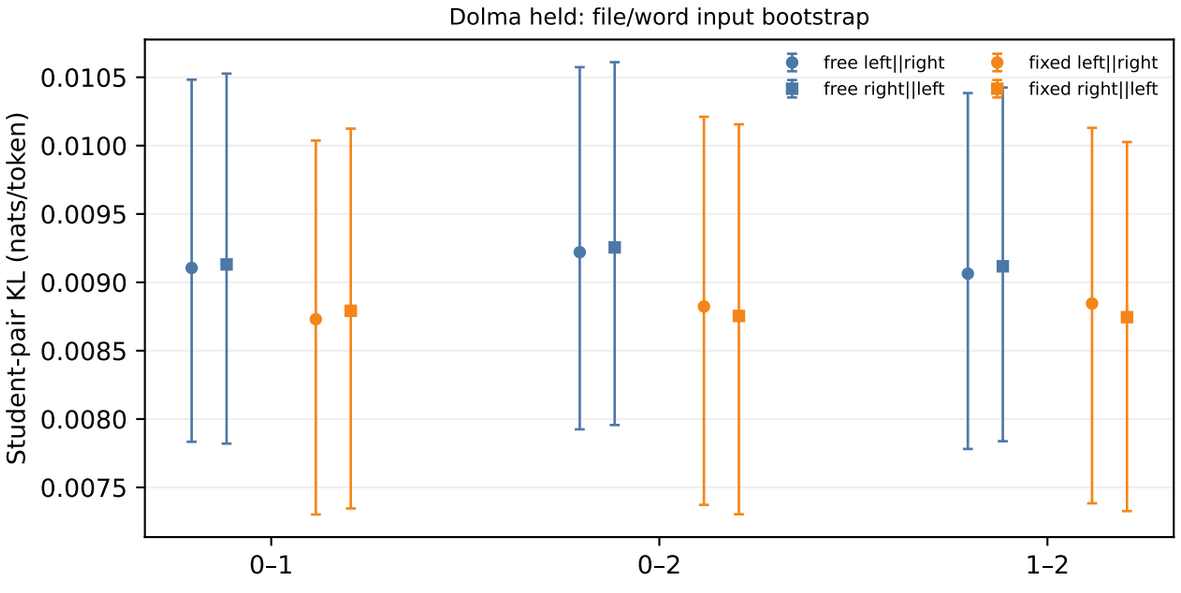
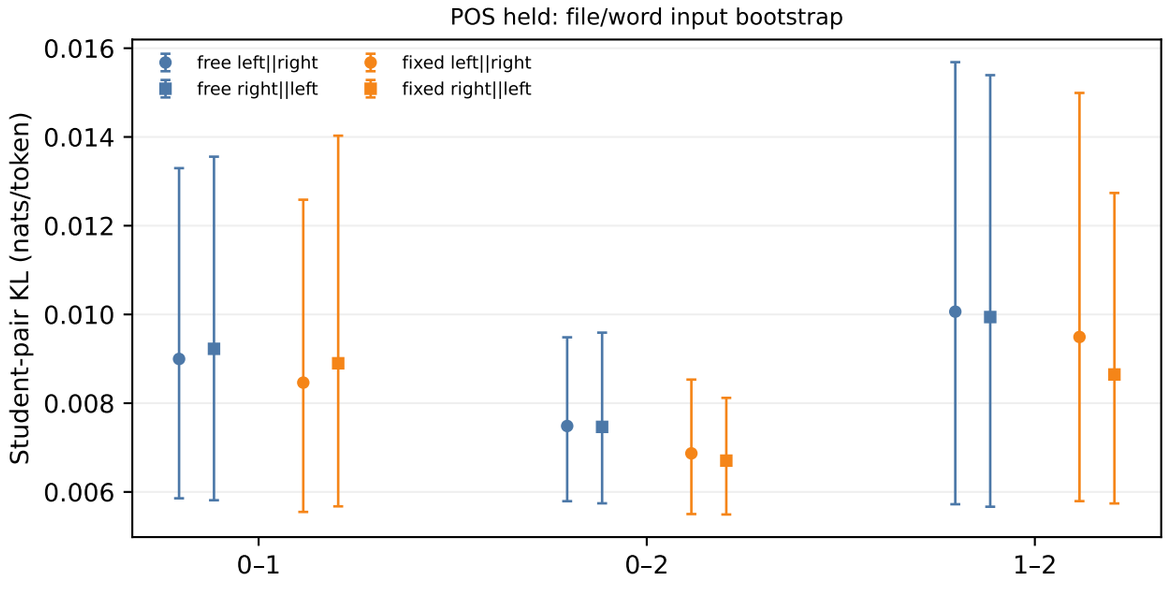
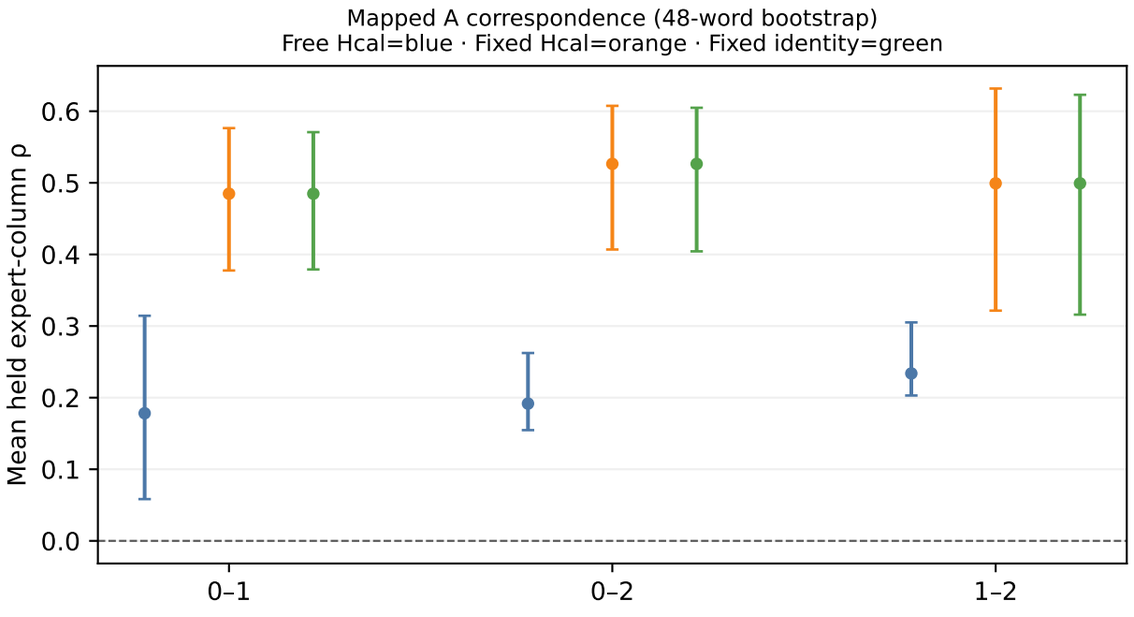
E139 — Student-pair natural KL and A







Scope

Free-map alignment is fitted once per state on training-only Hcal using seed1 as the registered anchor. Fixed Hcal maps use the same procedure; known identity is separate.

All intervals describe input-cluster uncertainty conditional on maps and the fixed three-student cohort, not seed uncertainty or uncertainty from refitting maps.

Free vs fixed changes the complete protocol (teacher routing information, frozen gate and related gradient/regularization coupling); it is not an isolated routing-cause estimate.

Dolma uses BF16 whole logits → FP32 log-softmax; POS uses BF16 backbone and genuine FP32 final head. CIs resample files/words, not student pairs or seeds.